

Program : Diploma in Tool & Die Engineering	
Course Code : 3117	Course Title: Engineering Metrology Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To introduce techniques for the precise measurement of various objects.
- To impart the basics of quality check & inspection methods in industries.

Course Prerequisites:

Topic	Course code	Course name	Semester
Introduction to Engineering measurements & metrology		Material Science & Metrology	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Measure various component of linear measurement using Vernier calipers and Micrometer	15	Applying
CO2	Measure various components of angle measurement using sine bar, bevel protractor, dial gauge	12	Applying
CO3	Calibrate the precision measuring instruments using slip gauge.	9	Applying
CO4	Interpret the application of different Gauges in quality check process.	6	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						3
CO2	3						3
CO3	3						3
CO4	3			3			3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Measure various component of linear measurement using Vernier calipers and Micrometer		
M1.01	Determine the thickness/depth of die plate/cavity insert to an accuracy of 0.02mm using Vernier caliper and check the result with digital/dial Vernier caliper.	3	Applying
M1.02	Determine the inside diameter of a bush component to an accuracy of 0.02 using Vernier caliper and check the result with digital/dial Vernier caliper.	3	Applying
M1.03	Determine the diameter of a guide pin to an accuracy of 0.01mm using micrometer and check the result with digital micrometer	3	Applying
M1.04	Determine the height of gauge block or parallel bars to an accuracy of 0.02mm using Vernier height gauge.	3	Applying
M1.05	Determine the depth of a blind bore component to an accuracy of 0.02mm using Vernier depth gauge.	3	Applying
CO2	Measure various components of angle measurement using sine bar, bevel Protractor, dial gauge.		
M2.01	Determine the angle of V-block, Taper Shank of Drill and Dovetails in mechanical components using universal bevel protractor.	6	Applying
M2.02	Determine the angle of machined surfaces of components using sine bar/Dial gauge with slip gauges.	6	Applying
	Lab Exam - I	1.5	

CO3	Calibrate the precision measuring instruments using slip gauge.		
M3.01	Calibrate the Vernier caliper/micrometer/dial gauge etc. using slip gauges.	9	Applying
CO4	Interpret the application of different Gauges in quality check process.		
M4.01	Apply plug gauge, ring gauge & snap gauges and perform quality checks.	3	Applying
M4.02	Summarize comparators and its applications.		Understanding
M4.03	Summarize CMM and its applications.		Understanding
M4.04	Inspect core & Cavity inserts using CMM	3	Applying
	Lab exam – II	1.5	

Text/ Reference:

T/R	Book Title/Author
T1	R.K.Jain., Engineering Metrology, Khanna Publishers, Eleventh edition, 1989
R1	R.Jenkins, Fundamentals of Mechanical Inspection, McGraw Hill Book company

Online Resources:

Sl.No	Website Link
1	https://easyengineering.net/engineering-metrology-and-measurements-by-raghavendra/
2	http://home.iitk.ac.in/~jrkumar/download/Lecture-4.pdf
3	https://www.academia.edu/34376633/ENGINEERING_METROLOGY_AND_MEASUREMENTS
4	https://westportcorp.com/blogs/precision-gages/types-of-mechanical-gauges
5	http://www.ignou.ac.in/upload/Unit-8-62.pdf